



Name:

Roll No:

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI

(An ISO 9001:2015 Certified College)

Affiliated to APJ Abdul Kalam Technological University

Second Internal Examination – December 2022

FIFTH SEMESTER – B. TECH (A&B)

CST 301: Formal Languages and Automata Theory

Time: 2 Hours.

Max.Marks:50

PART A (Answer All Questions)

Qn. No.	Questions	Marks	Question K level	CO & its K level
1	List any two closure properties of Context-Free Language	3	K2	CO3(K3)
2	With an example, explain ambiguity in Context-Free Grammar.	3	K2	CO3(K3)
3	List the application of the Myhill-Nerode Theorem	3	K2	CO2(K2)
4	Explain the different methods by which a PDA accepts a language.	3	K2	CO3(K3)
5	Write the formal definition of the Turing Machine.	3	K2	CO4(K3)

PART-B (Answer any 2 Questions)

Qn No.	Questions	Marks	Question K level	CO & its K level																					
6	State Myhill-Nerode Theorem, Obtain the minimum state DFA for the following DFA. <table><tr><td></td><td>a</td><td>b</td></tr><tr><td>→ 0</td><td>1</td><td>2</td></tr><tr><td>1</td><td>4</td><td>5</td></tr><tr><td>2</td><td>0</td><td>3</td></tr><tr><td>3</td><td>5</td><td>2</td></tr><tr><td>4</td><td>1</td><td>0</td></tr><tr><td>5</td><td>4</td><td>3</td></tr></table>		a	b	→ 0	1	2	1	4	5	2	0	3	3	5	2	4	1	0	5	4	3	7	K2	CO2(K2)
	a	b																							
→ 0	1	2																							
1	4	5																							
2	0	3																							
3	5	2																							
4	1	0																							
5	4	3																							
7	Design a PDA to accept the language $L = \{ww^R \mid w \in \{0,1\}^*\}$. Also, illustrate the computation of the PDA on the string 'aaaa'.	7	K3	CO3(K3)																					
8	Using the pumping lemma for Context-free languages, prove that the language $L = \{a^n b^n c^n \mid n \geq 1\}$.	7	K3	CO3(K3)																					

PART-C (Answer any 2 Questions)

9	Consider the following productions: $S \rightarrow aB \mid bA$ $A \rightarrow aS \mid bAA \mid a$ $B \rightarrow bS \mid aBB \mid b$ For the string 'baaabbba' find - The leftmost derivation - The rightmost derivation - The parse tree	7	K3	CO3(K3)
10	State and explain any three closure properties of Context-free Languages.	7	K2	CO3(K3)
11	Write the formal definition of Context Sensitive Grammar and write the CSG for the language $L = \{ a^n b^n c^n \mid n \geq 1 \}$. Also, write string derivation of $w = 'aaabbbccc'$.	7	K3	CO4(K3)

PART D (Mandatory Questions)

12	Design a Turing Machine that recognizes the language $L = \{ a^n b^n c^n \mid n > 0 \}$ Illustrate the computation of TM on the input 'aaabbbccc'.	7	K3	CO4(K3)
----	--	---	----	---------

Name:

Roll No:



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI

(An ISO 9001:2015 Certified College)

Affiliated to APJ Abdul Kalam Technological University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

First Internal Examination – November 2022

FIFTH SEMESTER – B. TECH (A&B)

Time: 2 Hours.

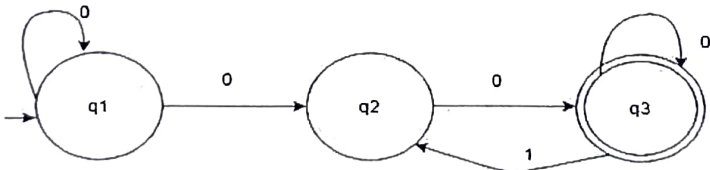
CST 301: Formal Languages and Automata Theory

Max.Marks:50

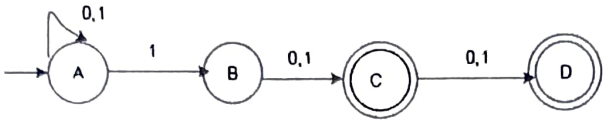
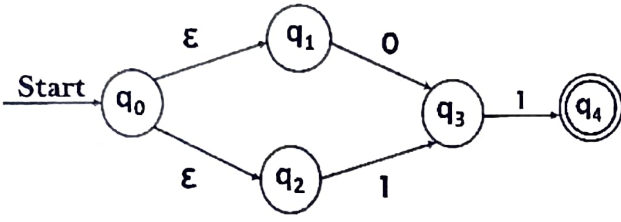
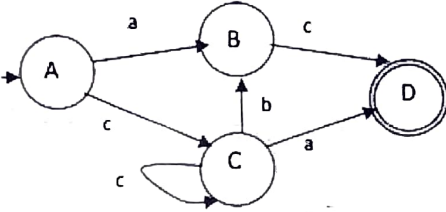
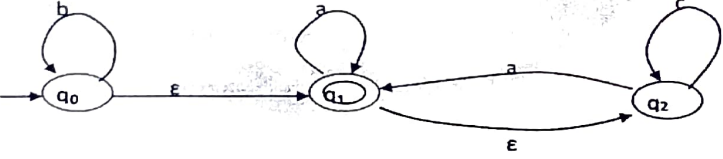
PART - A (Answer all Questions)

Qn. No.	Questions	Marks	K level of Qn.	CO & its K level
1	Differentiate between Positive Closure and Kleene Closure	3	K2	CO1(K2)
2	a.) Construct DFA for the following language; $L = \{\text{set of all strings that starts and ends with the same symbols, where } \Sigma = \{a,b\}\}$.	2	K2	CO2(K2)
	b.) Write the Formal definition of a DFA.	1		
3	Give a regular expression for the language: strings of '0' and '1' containing at least two '0'.	3	K2	CO2(K2)
4	Construct non-deterministic finite automata for regular expression $(0+1)^* 1$.	3	K2	CO2(K2)
5	Write the Formal definition of Grammar.	3	K1	CO2(K2)

PART - B (Answer any 2 Questions)

Qn No.	Questions	Marks	K level of Qn.	CO & its K level
6	Explain the different types of language classification.	7	K1	CO1(K2)
7	Convert the following NFA to DFA. 	7	K2	CO2(K2)
8	a.) What is the language is generated by given grammar? $S \rightarrow aS \mid bS \mid \epsilon$	3.5	K2	CO2(K2)
	b.) Differentiate between the transition function in DFA, NFA, and ϵ -NFA and explain the power of each, on the basis of language acceptability	3.5		

PART C (Answer any 2 Questions)

9	a.) Construct a regular expression corresponding to the following state diagram 	3.5		
	b.) Find the e-closure of all states of the automata. 	3.5	K2	CO2(K2)
10	Construct Regular grammar for the regular expression: $L = (a + b)(a + b)(a + b)^*$	7	K2	CO2(K2)
11	a). What is the regular expression for the DFA? 	3.5	K2	CO2(K2)
	b). Construct Regular grammar for the following language $L = \{a^m b^n \mid m \geq 1 \text{ and } n \geq 1\}$	3.5		
PART D (Mandatory)				
12	Convert the ϵ-NFA to NFA. 	7	K2	CO2(K2)

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: CST 301**Course Name: FORMAL LANGUAGES AND AUTOMATA THEORY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

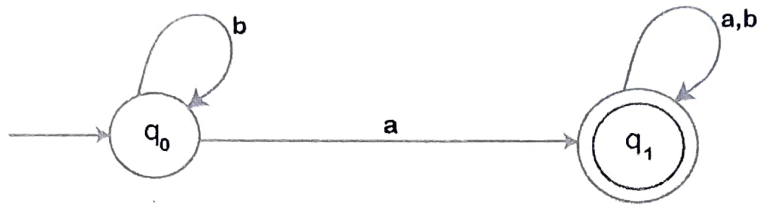
		Marks
1	Define ϵ -closure of a state? Explain with example.	3
2	Construct a DFA for strings in which first and last letters do not match. $\Sigma = \{a, b\}$	3
3	Generate regular expression for strings in which number of a's is a multiple of three. $\Sigma = \{a, b\}$	3
4	Explain any 3 closure properties of regular languages	3
5	With suitable example, explain about ambiguous grammar	3
6	State Myhill - Nerode Theorem	3
7	Whether DPDA and NPDA are equivalent? Justify your answer	3
8	Explain how CFGs can be converted to Chomsky Normal Form	3
9	Define Turing Machine	3
10	Differentiate between Recursive and Recursively Enumerable languages	3

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- 11 a) Prove that, if L is accepted by an ordinary NFA, there exist an equivalent c-NFA that also accepts L 7
- b) Design an NFA (without ϵ -moves) for strings having substring 'bab'. Convert it into equivalent DFA. $\Sigma = \{a, b\}$ 7
- 12 a) Construct an c-NFA for the language $L = \{0^n 1^m 2^p / n, m, p \geq 0\}$ and convert it into equivalent NFA without c-transitions 7
- b) Design an NFA (without ϵ -moves) for strings with either consecutive zeros or consecutive ones. Obtain its corresponding DFA 7

Module -2

- 13 a) Using pumping lemma, show that $L = \{a^n b^n / n > 0\}$ is not regular 7
 b) Develop equivalent automata for the Regular Expression $(a+b)^* aabb(a^*+bb)^*$ 7
- 14 a) Prove that for every Regular Expression 'R', there is an c-NFA 'M' 7
 b) List out the rules for writing regular expressions. Convert the following DFA to its equivalent Regular Expression

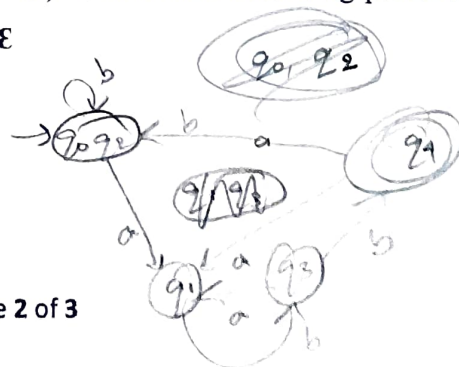
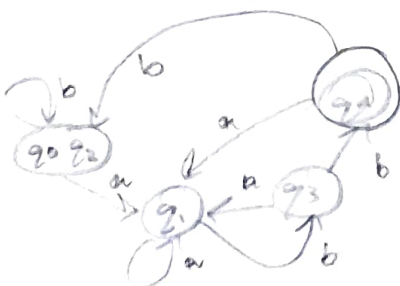


Module -3

- 15 a) What is Greibach Normal Form (GNF)? Convert the following CFG to GNF 7
 $S \rightarrow AA / a, \quad A \rightarrow SS / b$
- b) a) Design CFG for the following languages 7
 (i) Palindromes over $\{a, b\}$
 (ii) Stings with more than 2 zeros. $\Sigma = \{0, 1\}$
 (iii) $(0+1)^*(01)^*(0+1)^*$
- 16 a) Minimize the following DFA using Myhill – Nerode theorem

	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_1	q_3
q_2	q_1	q_2
q_3	q_1	$*q_4$
$*q_4$	q_1	q_2

- (b) What is Chomsky Normal Form (CNF)? Convert the following productions to CNF. 7
 $S \rightarrow aSa / bSb / SS / \epsilon$



Module -4

- 17 a) Prove that for every PDA accepted by final state, there exists an equivalent PDA accepted by empty stack. 7
- b) Design PDA for set of even length palindromes over $\{a, b\}$. Illustrate the working with suitable example 7
- 18 a) Design PDA for $L = \{x \in \{a, b\}^* / \#_a(x) = \#_b(x)\}$. 7
Here $\#_p(x)$ represents the number of occurrences of the symbol p in the string x
- b) Using pumping lemma for CFLs, show that $L = \{ww / w \in \{a, b\}^*\}$ is not context free. 7

Module -5

- 19 a) Design TM for $L = \{a^n b^m a^n / m, n > 0\}$. Illustrate the working with suitable example 7
- b) Explain Chomsky hierarchy for formal languages and evaluate various types 7
- 20 a) Design a TM to copy a string of a's and b's to the right side, leaving one blank symbol (b) in between. Assume that initially the input tape contains **xbx** and TM halts with **xbxbx** as the tape content. $x \in \{a, b\}^*$ 7
- b) Prove that TM halting problem is undecidable 7
